

Volume Probe Sweep Report

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Inputs

- /scratch/xl598/runs/llm-stratified/20260411_volume_probe_parallel/pixel_stratification_sweep
- /scratch/xl598/runs/llm-stratified/20260411_volume_probe_parallel/pretrained_pixel_sweep
- Aggregated JSON: /scratch/xl598/runs/llm-stratified/20260411_volume_probe_parallel/volume_probe_sweep_report.json

Coverage

sweep	completed_runs	incomplete_dirs
pixel_stratification_sweep	10	0
pretrained_pixel_sweep	8	0

Key Findings

- Overlapping raw patches increased irregularity in 1/4 matched dataset/patch pairs; the largest jump was STL10 patch 16 (0.861 -> 0.873).
- The most stratified raw-pixel configuration was STL10 patch 8 stride 8 (irregular_ratio=0.875), while the smoothest completed raw-pixel run was CIFAR10 patch 16 stride 8 (irregular_ratio=0.793).
- Across the pure pixel sweep, token irregularity was 0.052 higher than the matched raw-pixel probe on average (mean token-minus-pixel delta).
- DINO token representations were less irregular than their matched raw-pixel probes in 0/6 completed DINO runs (mean raw=0.852, mean token=0.900).
- In the multilayer DINO runs, irregularity decreased monotonically from layer 03 to 06 to last in 0/2 cases.
- Against the included untrained TinyViT baseline, DINO last-layer token irregularity was lower on 1/2 shared datasets.

Pixel Sweep

Metric cells use mean_dim / irregular_ratio.

dataset	patch	stride	raw_pixels	tokens	patch_embeddings
CIFAR10	4	4	8.18 / 0.833	4.57 / 0.949	7.98 / 0.832
CIFAR10	8	4	12.11 / 0.823	7.23 / 0.872	11.30 / 0.813
CIFAR10	8	8	11.36 / 0.825	7.23 / 0.872	11.30 / 0.813
CIFAR10	16	8	15.65 / 0.793	14.33 / 0.810	14.84 / 0.797
CIFAR10	16	16	15.23 / 0.807	14.33 / 0.810	14.84 / 0.797

dataset	patch	stride	raw_pixels	tokens	patch_embeddings
STL10	4	4	8.94 / 0.853	10.31 / 0.883	8.84 / 0.859
STL10	8	4	14.97 / 0.873	8.42 / 0.944	12.66 / 0.868
STL10	8	8	13.20 / 0.875	8.42 / 0.944	12.66 / 0.868
STL10	16	8	17.60 / 0.873	6.11 / 0.926	15.41 / 0.853
STL10	16	16	16.00 / 0.861	6.11 / 0.926	15.41 / 0.853

Pretrained Sweep

Metric cells use mean_dim / irregular_ratio.

dataset	variant	patch	stride	raw_pixels	patch_embeddings	tokens_or_last	layer_03	layer_06
CIFAR10	dinov2_last	14	7	4.90 / 0.838	10.36 / 0.807	5.10 / 0.906	n/a	n/a
CIFAR10	dinov2_last	14	14	5.32 / 0.833	10.36 / 0.807	5.10 / 0.906	n/a	n/a
CIFAR10	dinov2_multilayer	14	14	5.32 / 0.833	10.36 / 0.807	5.10 / 0.906	16.62 / 0.793	12.38 / 0.849
CIFAR10	untrained_tinyvit	8	8	11.36 / 0.825	11.30 / 0.813	7.23 / 0.872	n/a	n/a
STL10	dinov2_last	14	7	8.49 / 0.854	25.92 / 0.819	5.62 / 0.895	n/a	n/a
STL10	dinov2_last	14	14	10.41 / 0.877	25.92 / 0.819	5.62 / 0.895	n/a	n/a
STL10	dinov2_multilayer	14	14	10.41 / 0.877	25.92 / 0.819	5.62 / 0.895	22.57 / 0.808	15.58 / 0.830
STL10	untrained_tinyvit	8	8	13.20 / 0.875	12.66 / 0.868	8.42 / 0.944	n/a	n/a

Notes

- raw_pixels refers to the primary raw-pixel probe for that run: patch_pixels for non-overlapping patches or patch_pixels_stride_<k> when overlap was enabled.
- tokens_or_last maps to tokens for TinyViT runs and to the last-layer token representation for DINO runs.
- Comparisons across the DINO and untrained TinyViT rows are informative but not perfectly apples-to-apples because their patch grids differ.